

Homework 3

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R-4.7

A	B
$8n \log n = 2n^2$	
$4n \log n = n^2$	
$4 \log n = n$	
$4 = \frac{n}{\log n}$	

Therefore $n = 16$

$$\text{As, } 4 = \frac{16}{\log 16} = \frac{16}{4} = 4$$

Thus $n_0 = 17$ for A to be faster
since for all $n \geq 17$, A will be faster than B (because
if $n = 16$, A and B are equal)

R-4.18

The time complexity is $O(n^2)$.

R-4.8

A B

$$40n^2 = 2n^3$$

$$\frac{40n^2}{2} = \frac{2n^3}{2}$$

$$20n^2 = n^3$$

$$20 = n$$

Therefore $n_0 = 20$ for $n \geq 20$ or $n = 21$.

A will be faster than B as at $n = 20$, $A = B$

R-4.13

The order is given below:

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|---------------------------------|-----------------|
| 1. 2^{10} (constant) | 7. n^3 |
| 2. $4n$ | 8. $2^{\log n}$ |
| 3. $3n + 100 \log n$ (degree 1) | 9. 2^n |
| 4. $n \log n$ | |
| 5. $4n \log n + 2n$ | |
| 6. $n^2 + 10n$ | |
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